

Introducing Zearn Summer Math Intensive Series

Zearn's Summer Intensive Series is designed based on learnings of the billions of math problems completed on our math platform: where students struggle and the specific foundational support that kids need to access grade-level math content.

During the academic year, we identify and recommend these foundational lessons to teachers so they can be used for deeper interventions. Our Summer Series pulls forward these recommendations into coherent summer math experiences. Each Summer Intensive Series offers coherent, focused four-to-six-week sequences that build the strong foundations all rising 1st through 9th graders need to move forward during the upcoming academic year.

Comprehensive materials for 4-to-6-week programs

Each Series consists of top-rated materials that can be used flexibly across summer programs:

- ✓ **Daily digital math lessons** offer a consistent structure of learning activities, designed to accelerate math learning by integrating unfinished learning into the context of new learning. Students explore new concepts with real on-screen teachers, visualization of every math concept, interactive problem solving, and guided paper-and-pencil Student Notes.
- ✓ **Materials for hands-on problem solving** offer teachers and tutors daily application problems they can use to facilitate lively math discussions where students explore different ways to solve problems, unlocking creativity and joy with math.
- ✓ **Real-time reports on student learning** provide visibility into student learning and where students need additional support to move forward in their learning.

Focused on the big ideas of math

Each Summer Math Intensive Series deeply explores the essential math content that will help rising 1st–9th graders unlock grade-level math learning this fall. In younger grades, students explore one big idea, like addition and subtraction. In older grades, students tackle longer sequences covering a few big ideas, building a strong foundation in critical capstone content that they will need for success in later grades.

Recommended usage

In each 4-to-6-week program:



Rising 1st through 3rd graders should complete 1 lesson each day (~30 minutes)



Rising 4th through 9th graders should complete 2 lessons each day (~60 minutes)

Explore the content of Zearn's Summer Math Intensive Series on the following pages.

Summer Math Intensive Series: Rising 1st Graders

For rising 1st graders, the Kindergarten Summer Intensive Series focuses on the essentials of addition and subtraction to set students up for success in 1st grade.

Mission	Content for Rising 1st Graders: 33 Lessons			
	Standards Addressed	Mission Title	Lessons	Topics
GKM4	K.2I	Number Pairs, Addition and Subtraction to 10 This Mission marks the next exciting step in math for kindergartners—addition and subtraction! Students begin to harness their practiced counting abilities, knowledge of the value of numbers, and work with embedded numbers to reason about and solve addition and subtraction expressions and equations.	33	Topic A: Compositions and Decompositions of 2, 3, 4, and 5
	K.3A			Topic B: Decompositions of 6, 7, and 8 into Number Pairs
	K.3B			Topic C: Addition with Totals of 6, 7, and 8
	K.3C			Topic D: Subtraction from Numbers to 8
				Topic E: Decompositions of 9 and 10 into Number Pairs
				Topic F: Addition with Totals of 9 and 10
				Topic G: Subtraction from 9 and 10
				Topic H: Patterns with Adding 0 and 1 and Making 10

Summer Math Intensive Series: Rising 2nd Graders

For rising 2nd graders, the Grade 1 Summer Intensive Series focuses on counting, composing, and decomposing numbers. The Grade 1 Summer Intensive Series is intended to build a strong foundation of flexible strategies for all students to fall back on as they engage with Grade 2 content and beyond.

Mission	Content for Rising 2nd Graders: 32 Lessons			
	TEKS Standards Addressed	Mission Title	Lessons	Topics
<u>G1M1</u>	1.3B	Add and Subtract Small Numbers In this Mission, students make significant progress towards fluency with addition and subtraction of numbers to 10 as they are presented with opportunities intended to advance them from counting all to counting on, which leads many students then to decomposing and composing addends and total amounts. In Kindergarten, students achieved fluency with addition and subtraction facts to 5. This means they can decompose 5 into 4 and 1, 3 and 2, and 5 and 0. They can do this without counting all. They perceive the 3 and 2 embedded within the 5.	32	Topic A: Embedded Numbers and Decompositions
	1.3D			Topic B: Counting On from Embedded Numbers
	1.3E			Topic C: Addition Word Problems
	1.5D			Topic D: Strategies for Counting On
	1.5E			Topic E: The Commutative Property of Addition and the Equal Sign
	1.5G			Topic F: Development of Addition Fluency Within 10
				Topic G: Subtraction as an Unknown Addend Problem
				Topic H: Subtraction Word Problems
				Topic I: Decomposition Strategies for Subtraction
				Topic J: Development of Subtraction Fluency Within 10

Summer Math Intensive Series: Rising 3rd Graders

For rising 3rd graders, the Grade 2 Summer Intensive Series focuses on the capstone content of K–2: a deep understanding of the base ten system and flexibly being able to add and subtract. This is content that 3rd-grade standards assume students can access.

	Content for Rising 3rd Graders: 20 Lessons			
Mission	TEKS Standards Addressed	Mission Title	Lessons	Topics
<u>G2M5</u>		Add and Subtract Big Numbers	20	Topic A: Strategies for Adding and Subtracting Within 1,000
	2.4B	Students are now well on their way to mastering flexible addition and subtraction. This Mission builds on those skills, now using numbers up to 1,000 and increasing the focus on efficiency and checking their work.		Topic B: Strategies for Composing Tens and Hundreds Within 1,000
	2.4C			
	2.4D			Topic C: Strategies for Decomposing Tens and Hundreds Within 1,000
	2.7B			
	2.7C			Topic D: Student Explanations For Choice of Solution Methods

Summer Math Intensive Series: Rising 4th Graders

For rising 4th graders, the Grade 3 Summer Intensive Series focuses on multiplication and division of whole numbers and the understanding of fractions as numbers. Both of these big ideas are key to students' success in later grades.

Content for Rising 4th Graders: 55 Lessons				
Mission	TEKS Standards Addressed	Mission Title	Lessons	Topics
<u>G3M3</u>	3.4E	Multiply and Divide Tricky Numbers This Mission extends multiplication and division to all factors between 0 and 10. Students work deeply with the commutative, distributive, and associative properties and have problem-solving opportunities in each topic.	18	Topic A: Multiplication as Comparison
	3.4F			Topic B: The Properties of Multiplication and Division
	3.4G			Topic C: Division as an Unknown Factor Problem
	3.4I			Topic D: Multiplication and Division Using Units up to 8
	3.4J			Topic E: Multiplication and Division Using Units of 9
	3.4K			Topic F: Analysis of Patterns and Problem Solving Including Units of 0 and 1
	3.5A			Topic G: Multiplication of Single-Digit Factors and Multiples of 10
	3.5B			
	3.5C			
	3.5D			
	3.5E			
<u>G3M4</u>	3.6C	Find the Area In this Mission, students explore area as an attribute of two-dimensional figures and relate it to their prior understandings of multiplication.	8	Topic A: Concepts of Area Measurement
	3.6D			Topic B: Arithmetic Properties Using Area Models
<u>G3M5</u>	3.3A	Fractions as Numbers This Mission builds a deep understanding of fractions as a single number. Students begin concretely interacting with fractions by folding strips of paper to create equal parts. By the end of the Mission, students will be able to evaluate equivalence and compare size pictorially.	29	Topic A: Partitioning a Whole into Equal Parts
	3.3B			Topic B: Unit Fractions and their Relation to the Whole
	3.3C			Topic C: Comparing Unit Fractions and Specifying the Whole
	3.3D			Topic D: Fractions on the Number Line
	3.3E			Topic E: Equivalent Fractions
	3.3F			Topic F: Comparison, Order, and Size of Fractions
	3.3G			
	3.3H			
	3.4K			
	3.5A			
	3.5B			
	3.6E			
	3.7A			

Summer Math Intensive Series: Rising 5th Graders

For rising 5th graders, the Grade 4 Summer Intensive Series focuses on fractions and decimals. Students evaluate equivalence and learn to extend that understanding to decimal operations. The Summer Intensive Series is intended to ensure students have a solid grasp of these ideas to close out 5th grade successfully.

Content for Rising 5th Graders: 45 Lessons				
Mission	TEKS Standards Addressed	Mission Title	Lessons	Topics
<u>G4M5</u>	4.3A	Equivalent Fractions This Mission teaches students how to manipulate fractions. Students compare fractions, evaluate equivalence, and learn that the same methods they used for whole number operations can be used to add and subtract fractions.	29	Topic A: Decomposition and Fraction Equivalence
	4.3B			Topic B: Fraction Equivalence Using Multiplication and Division
	4.3C			Topic C: Fraction Comparison
	4.3D			Topic D: Fraction Addition and Subtraction
	4.3E			Topic E: Extending Fraction Equivalence to Fractions Greater Than 1
	4.3F			Topic F: Addition and Subtraction of Fractions by Decomposition
	4.3G			
	4.8C			
	4.9A			
	4.9B			
<u>G4M6</u>	4.2A	Decimals Students learn a new and special notation for fractions in this Mission: decimals! They extend their understanding of the base ten system by first exploring equivalence between fractions and decimals. This understanding extends to comparing decimals and adding money.	16	Topic A: Exploration of Tenths
	4.2B			Topic B: Tenths and Hundredths
	4.2E			Topic C: Decimal Comparison
	4.2F			Topic D: Addition and Subtraction with Tenths and Hundredths
	4.2G			Topic E: Money Amounts as Decimal Numbers and Financial Literacy
	4.2H			
	4.3E			
	4.4A			
	4.8C			
	4.10A			
	4.10B			
	4.10C			
	4.10D			
	4.10E			

Summer Math Intensive Series: Rising 6th Graders

For rising 6th graders, the Grade 5 Summer Intensive Series focuses on essential operations with whole numbers, fractions, and decimals. The Summer Intensive Series is intended to ensure students are prepared for 6th-grade math as an extension of elementary school work.

Content for Rising 6th Graders: 55 Lessons				
Mission	TEKS Standards Addressed	Mission Title	Lessons	Topics
<u>G5M2</u>	5.2A	Base Ten Operations Mission 1 introduced students to decimal fractions, and Mission 2 will introduce them to similar conceptual work (mental math, reasoning, conceptual models, and algorithms), but with whole numbers. Students will multiply multi-digit whole numbers in the first part of the Mission and divide them in the end. Each operation concludes with application work and measurement word problems.	29	Topic A: Prime and Composite Numbers
	5.3A			Topic B: Mental Strategies for Multi-Digit Whole Number Multiplication
	5.3B			Topic C: The Standard Algorithm for Multi-Digit Whole Number Multiplication
	5.3C			Topic D: Decimal Multi-Digit Multiplication
	5.3D			Topic E: Measurement Word Problems with Whole Number and Decimal Multiplication
	5.3E			Topic F: Mental Strategies for Multi-Digit Whole Number Multiplication
	5.3F			Topic G: Partial Quotients and Multi-Digit Whole Number Division
	5.3G			Topic H: Partial Quotients and Multi-Digit Decimal Division
	5.4A			Topic I: Measurement Word Problems with Multi-Digit Division
	5.4B			
	5.4E			
	5.4F			
	5.7A			
<u>G5M3</u>	5.3A	Add and Subtract Fractions In this Mission, students will develop flexibility with addition and subtraction of fractions so they can mentally or numerically solve, reason, and estimate their calculations. The Mission begins with concrete and pictorial work (using area models and number lines) and moves to numeric work with word problems.	13	Topic A: Making Like Units Pictorially
	5.3H			Topic B: Making Like Units Numerically
	5.3K			Topic C: Further Applications

G5M4	5.3I	Multiply and Divide Fractions Students tackled equivalency in adding and subtracting fractions in Mission 3; now they're ready to focus on multiplying and dividing fractions. Their understanding will be deeply rooted in multiple concrete examples and pictures before they generalize to more abstract methods.	13	Topic A: Repeated Addition of Fractions as Multiplication
	5.3J			Topic B: Multiplication of a Whole Number by a Fraction
	5.3L			Topic C: Fraction Expressions and Word Problems
	5.4E			Topic D: Division of Fractions
	5.4F			Topic E: Applying Fraction Multiplication to Personal Financial Literacy
	5.7A			Topic F: Interpretation of Numerical Expressions
	5.9A			
	5.9C			
	5.10A			
	5.10B			
	5.10C			
	5.10D			
	5.10E			
	5.10F			

Summer Math Intensive Series: Rising 7th Graders

For rising 7th graders, the Grade 6 Summer Intensive Series begins by developing understanding of ratios and rates so that students can represent and think about them in multiple, flexible ways. This is important because these concepts will be the foundations of proportional relationships and linear equations. The Summer Intensive Series also includes additional Missions that focus on bridging arithmetic from 3rd to 5th grade to 7th-grade arithmetic that requires students to deeply understand and have a higher fluency with operations.

Content for Rising 7th Graders: 44 Lessons				
Mission	TEKS Standards Addressed	Mission Title	Lessons	Topics
<u>G6M3</u>	6.4D	Unit Rates and Percentages In this Mission, students extend their developing understanding of ratios to dig deeper into ratio relationships and learn about rates and unit rates. Students are introduced to the idea of a percentage, which allows them to solve a variety of real-world problems.	14	Topic A: Using Unit Rate to Solve Problems
	6.4F			Topic B: Unit Conversion
	6.4G			Topic C: Rates
	6.4H			Topic D: Percentages
	6.5A			Topic E: Let's Put It to Work
	6.5B			
<u>G6M4</u>	6.3A	Dividing Fractions In this Mission, students culminate their study of fractions as they learn to divide a fraction by a fraction, using concrete examples and real-world contexts to help them make sense of the mathematics.	16	Topic A: Making Sense of Division
	6.3E			Topic B: Meanings of Fraction Division
	6.8C			Topic C: Algorithm for Fraction Division
	6.8D			Topic D: Fractions in Lengths, Areas, and Volumes
				Topic E: Let's Put It to Work
<u>G6M5</u>	6.3E	Arithmetic in Base Ten In this Mission, students use various models and representations to add, subtract, multiply, and divide decimals and formalize their work with the appropriate standard algorithms.	14	Topic A: Warming up to Decimals
	6.7C			Topic B: Adding and Subtracting Decimals
				Topic C: Multiplying Decimals
				Topic D: Dividing Decimals
				Topic E: Let's Put It to Work

Summer Math Intensive Series: Rising 8th Graders

For rising 8th graders, the Summer Intensive Series focuses on three big ideas of Grade 7: proportional relationships, arithmetic with signed numbers and solving multi-step equations. Proportional relationships are revisited and extended in 8th grade to introduce students to the concept of a function, which is arguably the most important concept students will study for the remainder of their mathematical careers. Additionally, students' work with rational number arithmetic and solving multi-step equations is woven throughout 8th grade, requiring students to have a certain degree of fluency with each to allow their working memory to focus on the new learning of Grade 8.

	Content for Rising 8th Graders: 51 Lessons			
Mission	TEKS Standards Addressed	Mission Title	Lessons	Topics
<u>G7M4</u>	7.4A 7.4B 7.4C	Proportional Relationships and Percentages In this Mission, students use ratios, scale factors, unit rates, and proportional relationships to solve multi-step, real-world problems that involve fractions and percentages. Students represent amounts and corresponding percent rates with double number line diagrams and tables.	14	Topic A: Proportional Relationships with Fractions Topic B: Percent Increase and Decrease Topic C: Applying Percentages Topic D: Let’s Put It to Work
<u>G7M5</u>	7.3A 7.3B 7.4A	Rational Number Arithmetic In this Mission, students use tables and number line diagrams to represent sums and differences of signed numbers or changes in quantities represented by signed numbers such as temperature or elevation, becoming more fluent in writing different numerical addition and subtraction equations that express the same relationship.	14	Topic A: Interpreting Negative Numbers Topic B: Adding and Subtracting Rational Numbers Topic C: Multiplying and Dividing Rational Numbers Topic D: Four Operations with Rational Numbers Topic E: Solving Equations Where There are Negative Numbers Topic F: Let’s Put It to Work
<u>G7M6</u>	7.10A 7.10B 7.10C 7.11A 7.11B	Expressions, Equations, and Inequalities In this Mission, Students learn algebraic methods for solving equations. Students solve linear inequalities in one variable and represent their solutions on the number line. They understand and use the terms “less than or equal to” and “greater than or equal to,” and the corresponding symbols.	22	Topic A: Representing Situations of the Form $px + q = r$ and $p(x + q) = r$ Topic B: Solving Equations of the Form $px + q = r$ and $p(x + q) = r$ and Problems That Lead to Those Equations Topic C: Inequalities Topic D: Writing Equivalent Expressions

Summer Math Intensive Series: Rising 9th Graders

For rising 9th graders, the Summer Intensive Series focuses on linear functions and linear equations, ensuring students have both the foundational understanding and procedural fluency needed to access the new learning of Algebra I. Algebra I is largely the study of functions that behave similarly. Linear functions serve as the launch point for the course. Students need a deep understanding of linear relationships and functions in addition to being able to manipulate algebraic equations and expressions, making these topics the natural choice for the Grade 8 Summer Intensive Series.

Content for Rising 9th Graders: 49 Lessons				
Mission	TEKS Standards Addressed	Mission Title	Lessons	Topics
G8M3	8.4A	Linear Relationships In this Mission, students deepen their understanding of slope and learn to recognize connections among rate of change, slope, and constant of proportionality, as well as between linear and proportional relationships.	13	Topic A: Proportional Relationships
	8.4B			Topic B: Representing Linear Relationships
	8.5E			Topic C: Finding Slopes
	8.8A			Topic D: Linear Equations
	8.10A			Topic E: Let's Put It to Work
G8M4	8.8A	Linear Equations and Systems In this Mission, students write and solve linear equations in one variable. These include equations in which the variable occurs on both sides of the equal sign, and equations with no solutions, exactly one solution, and infinitely many solutions.	14	Topic A: Puzzle Problems
	8.8B			Topic B: Linear Equations in One Variable
	8.8C			Topic C: Systems of Linear Equations
	8.9A			Topic D: Let's Put It to Work
G8M5	8.4C	Functions and Volume In this Mission, students are introduced to the concept of a function. They describe functions as increasing or decreasing between specific numerical inputs, and they consider the inputs of a function to be values of its independent variable and its outputs to be values of its dependent variable.	18	Topic A: Inputs and Outputs
	8.5B			Topic B: Representing and Interpreting Functions
	8.5G			Topic C: Linear Functions and Rates of Change
	8.5H			Topic D: Cylinders and Cones
	8.5I			Topic E: Dimensions and Shapes
	8.6A			Topic F: Let's Put It to Work
	8.6B			
	8.7A			